Maheshwarachari Eshwarachari

Professional Summary

Telecom Testing & Automation/Performance Domain Expert with 12+ years of specialized expertise in telecommunications systems testing, automation framework development, DevOps practices, and performance engineering. Proven track record as Automation Lead & Test Lead architecting enterprise-scale test solutions across 2G/3G/4G/5G networks. Demonstrated expertise in designing comprehensive automation strategies, leading QA teams, implementing CI/CD pipelines, and delivering high-performance, mission critical telecom applications. Specialized in test automation, automation framework architecture, test leadership, performance testing, load testing, and DevOps infrastructure for telecom domain.

Domain Expertise for Target Roles:
Telecom Testing: 12+ years hands-on experience in 2G/3G/4G/5G network testing, OSS/BSS systems, billing validation, and subscriber lifecycle management

Test Automation & Automation Lead: Architected different automation frameworks like WAF, ENTSTAF, Xtaf achieving 85% test automation coverage; led automation teams; designed reusable test frameworks

Test Lead & QA Leadership: Spearheaded QA strategy for major releases; managed SIT (System Integration Testing) for complex telecom projects; mentored 15+ QA engineers

Performance & Load Testing: Architected load testing suites supporting 10,000+ concurrent users; implemented K6-based performance frameworks; built automated KPI dashboards

DevOps Engineering: Led DevOps transformation reducing deployment by 40%; architected Jenkins CI/CD pipelines; designed Terraform IaC; optimized AWS/Kubernetes infrastructure

Category	Tools
Programming Languages	Python, Java, JavaScript
Test Automation Frameworks	Robot Framework, Cypress Playwright, Karate
Performance / Load Testing Tools	K6, JMeter, Blaze Meter
Telecom Protocol Simulators	Seagull, Spirent
Network Analysis Tools	Wireshark, Tomcat
API Testing Tools	Postman, SoapUI
Cloud	Kubernetes, AWS, Azure, Jenkin

Contact

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ONSITE-BUSINESS TRAVELS

- Travelled to Austria in 2019 on a business travel
- Travelled to France in 2022 on a business travel
- Holding business visa for US, Argentina
- Worked in UK from 2022 to till date

Certifications

Experience

Senior Automation Engineer – June/2023 - Current
Capgemini UK PLC – Newbury, United Kingdom
Project: Vodafone [ENTS]
Project details:
European Network Testing Service (ENTS), a Vodafone-managed lab services organisation that oversees network testing across Europe while Capgemini handles day-to-day lab operations.

Roles and Responsibilities:
Automation:

Full-Stack Automation Framework: Designed and implemented comprehensive automation framework using Cypress, Karate, playwright for both API and GUI automation, integrating K6 for frontend and backend performance testing.

Multi-Protocol Automation Suite: Built Python + Robot Framework based automation suite covering GUI, API, and tool/plugin-based automation (Seagull, Spirent) to simulate Diameter and Radius protocol traffic for telecom systems

Network Node Automation: Developed end-to-end automation framework for 4G, 5G, and IMS network nodes, enabling scalable, reusable, and CI/CD-ready test execution

4G Core Network Nodes: Automated comprehensive 4G core network testing including PGW, SGW, MME, HLR with full functional and performance validation

5G Core Network Nodes: Architected automation for advanced 5G infrastructure nodes including AMF, SMF, UPF, AUSF, UDM, NRF, PCF with real-time protocol validation and with daily health checks

IMS Nodes Automation: Implemented automated testing for IMS infrastructure nodes including ASBG, CSCF, vBGF

Protocol-Level Automation: Developed automation libraries for validating 4G, 5G, and IMS protocols including Diameter, Radius, SIP, GTP-C, GTP-U, HTTP/2 and telecom signalling protocols

Protocol Simulator Integration: Integrated protocol simulators (Seagull, Spirent) into CI pipelines enabling high-volume signalling traffic simulation and validation

Performance & Reliability Engineering: Implemented Jmeter, Blaze Meter, K6-based performance test suites for API, backend microservices, and UI flows with automated dashboards and reporting

Performance Metrics & KPIs: Built comprehensive dashboards tracking latency, throughput, and load-based KPIs with automated reporting and SLA validation

Senior Test Analyst – May/2022 – June/2023
Capgemini UK PLC – Newbury, United Kingdom
Project: NOKIA [WING], Vodafone [ENTS]
Project details:

- Altran certification in python for python developers
- Certified SAFe 5 practioner, powered by scaled agile
- Performance Testing with JMeter for absolute beginners from course era

Education

MASTER OF COMPUTER APPLICATIONS
Cambridge Institute Of Technology

BACHELOR OF COMPUTER APPLICATIONS
Reva University

XII, **Science**, **2010**
S.B.G.N.S Rural Composite PU - College, Karnataka - Bangalore

SSLC FROM, **Science**, **2007**
Sir M V M High School - Chikkaballapur

Apreciations

- Appreciation mail for raising more than 70 valid BUGs in a span of 4 months in 2017
- Have been recognized under WOW - Team Excellence Award for Quality Delivery in 2017
- Have been recognized under WOW - Team Excellence Award for Quality Delivery in 2018
- Have been recognized under WOW – Individual Excellence award for Valuable contribution to team in 2020
- Have been recognized under WOW – Team Excellence Award for quality delivery in 2021

Languages

- Speak ENGLISH, KANNADA, TELUGU, TAMIL, HINDI
- Read ENGLISH, KANNADA, TELUGU, HINDI
- Write ENGLISH, KANNADA, TELUGU, HINDI

C2 General

The NOKIA WING is the worldwide IOT solution provided by Nokia for MVNO to become fully managed MVNO and to launch their services globally for their enterprise customers. WING provides the customer with the BSS system (CMP) and the entire 3G/4G core solution so that the subscriber can latch on the existing radio network of the MVNO and via Nokia WING core to get services like 3G, 4G data and SMS and there are other various features provided along with this.

Roles and Responsibilities:

Automation:

- Possess extensive experience in designing and customizing automation tools from scratch, including the development of WAF (Wing Automation Framework) for GUI, API, and mobile device automation, utilizing a range of tools and technologies such as Robot Framework, Python, Selenium, Appium, Jenkins, and Tomcat
- Extensive experience in architecting and developing end-to-end automation frameworks for 4G and 5G telecom network nodes, covering EPC and 5GC components such as MME, SGW/PGW, HSS, PCRF, AMF, SMF, UPF, and IMS nodes (CSCF, MRF, MGCF).
- Skilled in automating key telecom protocols—including SIP, Diameter, GTP-C/U, HTTP/2, NAS, SCTP, and RTP—by building custom, scalable test harnesses and traffic simulation tools.
- Have implemented automation solutions leveraging Python, Robot Framework, and CI/CD pipelines using Jenkins, GitLab/GitHub.
- Developed and implemented end-to-end automation framework and test suites, leveraging agile development methodologies and continuous integration and delivery pipelines, to ensure rapid and accurate testing of complex software systems
- Hands-on experience with JavaScript, Python, Linux, and bash scripting, leveraging automation capabilities to streamline testing processes and enhance efficiency.

Devops/Cloud:

- Proficient in writing comprehensive Terraform scripts to define infrastructure as code, ensuring reproducibility and scalability.
- In-depth familiarity with various cloud services and tools offered by AWS, enabling effective utilization and optimization for project requirements.
- Proficient in crafting Ansible playbooks to seamlessly provision, configure, and deploy complex applications and infrastructure.
- Demonstrated expertise in executing Git commands and employing efficient workflows, including branching, merging, and conflict resolution.
- Proficient in implementing Jenkins pipeline as code, facilitating continuous integration and delivery of software projects.

Web services:

- Possess extensive experience in testing web services, including both RESTful and SOAP services, using industry-standard tools such as JMeter, SoapUI, Postman, Swagger and have good experience in conducting both functional and non-functional testing like [performance, Load, Security]

- I am skilled in developing test plans, test cases, and test scripts for web services testing [black box/white box testing], and boundary value analysis, to ensure comprehensive test coverage

Team Leading:

- Spearheaded the planning and execution of system Devops, Infrastructure as code, CICD pipelines, integration testing (SIT), unit testing, automation testing, load testing, and performance testing for Nokia WING Releases, ensuring compliance with quality standards and adherence to established software development methodologies.
- Proficient in Software Testing Life Cycle (STLC) methodologies and experienced in Agile, V&V, and Waterfall Test environments.
- Skilled in deriving test plans and strategies for newly added functionalities, utilizing a range of techniques such as risk-based testing and exploratory testing to ensure effective testing coverage

Performance Testing:

- Designed and executed performance and load test scripts [Soap and Rest API's] using range of tools including JMeter, Blaze meter to measure system performance and load under various workloads and conditions
- Experience in developing a comprehensive, customized performance and load evaluation test suite/framework, utilizing a free range of tools such as JMeter, Blaze Meter [. JMX], Jenkins, Tomcat, and Python.
- Proficient in utilizing tools like ELK, Splunk, for monitoring during test execution, to ensure optimal performance and to identify areas of improvement.
- Skilled in running various load testing strategies, including stress testing, endurance testing, Soak testing and spike testing, to identify system bottlenecks and ensure the scalability and reliability of complex software systems
- Skilled in analyzing test results and collaborating with development teams to perform root cause analysis of performance issues, ultimately leading to the identification and implementation of corrective actions.

Customer Centric:

- Demonstrated experience in delivering successful customer demos and presentations
- Proven experience in managing customer projects, including project planning, execution, and delivery, while ensuring customer satisfaction and meeting project deadlines
- Demonstrated ability to work closely with customers to identify and prioritize project requirements, and to communicate project status and progress effectively
- Understanding of project management methodologies such as Agile, Waterfall, and Scrum, and the ability to apply those methodologies to customer projects as needed

End to End execution:

C2 General

- Proficient in reporting bugs, risks, and issues to the project team, ensuring effective communication and collaboration to drive prompt resolutions.
- Experienced in conducting test triage with technical architects, development leads, and platform support, facilitating comprehensive analysis of results and aligning stakeholders on the necessary actions.
- Contributed to the creation and publishing of relevant reports, including status updates and completion reports, to project stakeholders, ensuring transparency and keeping all parties informed about the project's progress.
- Proficient in onboarding accounts, subscriptions, and execution of DATA, VOICE, SMS sessions via real sims/simulators, including Sigos, Seagull, Spirent, CDBS OCS/RESTGW simulator, and performing billing functionalities checks.
- Possess expertise in node-level testing, with strong analytical and debugging skills in HLR/HSS, PGW, DRA, AAA, and RESTGW.
- Work experience with microservices, Kubernetes, and strong knowledge of SQL and database management systems (MySQL), capable of designing and implementing scalable and highly available database architectures

Senior Consultant I – Jan/2018 – May/2022
Capgemini – Bangalore, India
Project: NOKIA WING
Roles and Responsibilities:

- SIT Lead for Nokia WING Release testing
- Involved in Performance and load testing
- Contributed in developing Framework for automating provision applications and mobile device in robot frame work with python, Appium, selenium and for Performance & load testing tool integration to suite requirement
- Deriving test plan for newly added functionalities
- Peer review of test cases and involved in execution of API test case using JSON with the help of SOUP/REST/POSTMAN clients
- Regression testing for every new build
- Complete end to end Rest API testing and automation of Rest API in robot framework with python
- Performance and load testing using Jmeter, Webload, Blaze meter Provisioning of mobile subscribers and applying offers
- Execution of the CS and PS Call's and checking the Billings functionalities
- Timely QC update of executed test cases
- Reporting and tracking issues using QC
- Regular project review meetings with internal teams and with Customer

Senior Test Engineer – Feb/2017 – Jan/2018
Capgemini – Bangalore, India
Project: Motorola [Public safety application]
Project details:
Motorola's UPP (Unified PTT Product) is an IMS Based LTE standard Push to Talk over cellular System which provide Half duplex communication over packet data network (3g and 4G). The system working model is based on standard IMS Architecture with UPP server having all the inbuilt capability to function as HSS, CSCF, P-CSCF, S-CSCF, MRF, PSTN Gateway (to IDEN n/w) and Charging. The system allows us to make VOIP calls using the Android/IOS phones having the

SIP clients (W7K APP) Installed on it. The Entire Network's Operation and Maintained is carried out by Linux based OMC from where all the NE's Upgrade/Fallback, Disaster recovery, Alarms/events, PM Stats and other NE related operations are taken care.

Roles and Responsibilities:

- Understanding user requirements and system specifications
- Deriving test cases for new functionality
- Test Bed setup and configuration
- Involved in Execution of test case, and peer Review of test cases
- Regression testing for every new build
- Scratch installation and system upgrade (includes Installation of EXSi vSphere Client in HP DL380 G8 and G9 servers) + Hands on experience for RPM package installation
- Execution of the SIP based Call processing on Functional test cases like Feature, Regression and Sanity
- Positive and negative testing for client application validation for IOS and androide mobile applications
- Involved in Setup migrations/upgrade, and integrating the setups with IMW (intellectual Middle ware)
- Working with VMware virtual machines
- Lab auditing and processing of the observed data to into reports
- Daily observations of Lab machines, lab servers and lab equipment
- Upgrading OS for IOS/androide mobile devices and analyzing logs generated by the device/application (adb, tcpdump, application logs)
- Reporting and tracking issues using HP ALM, JIRA
- Regular project review meetings with internal teams and with Customer
- Writing/Running automation scripts using python in google Appium framework

Senior Test Engineer – March/2016 – Feb/2017
Capgemini[Aricent, Altran] – Bangalore, India
Project: Amadocs [C1 Solution]
Roles and Responsibilities:

- Understanding user requirements and system specifications
- Deriving test cases for new functionality + Involved in Execution of test case, and peer Review of test cases
- Unit testing, and system testing
- Basic knowledge on Offline billing and Online Charging System (CSM,
- CRM, RC, and NRC)
- Involved in Functionality (GUI) and Integration testing
- Regression testing for every new build
- Reporting and tracking issues using IBM Clear-Quest
- Analyzing logs generated by the application
- Regular project review meetings with internal teams and with Customer

Associate Trainee, Associate Test Engineer, **March/2013 – Feb/2016**

- Lab auditing and processing of the observed data to into reports
- Daily observations of Lab machines, lab servers and lab equipment
- Worked on OSS/BSS products (Mediation, Web Service Applications) + Involved in GUI testing and functional testing

- Regression testing for every new build
- Web services testing using Soap UI and knowledge on WSDL
- Scratch installation and system upgrade
- Hands on experience for RPM package installation
- Reporting and tracking issues using Defect Tracking Tool - Bugzilla
- Positive and negative testing for validation
- Analyzing the Linux and java log files generated by the application